

MK
KS
ML
AB
HT
CF
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GS
AM/YR
DH
WS

NAME _____

CLASS 12MT____ **or 12MTX**____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2018

YEAR 12

AP4

MATHEMATICS

Time allowed – 3 HOURS + 5 Minutes Reading Time

DIRECTIONS TO CANDIDATES:

SECTION I: Use the multiple-choice answer sheet provided.

SECTION II:

- Each question is to be commenced in a new booklet clearly marked Question 11, Question 12, etc on the top of the page.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- If you do not attempt a question, you must submit a blank booklet clearly indicating the question number, your name and class
- All questions should be placed in order - multiple-choice answer sheet followed by Question 11 to 16.
- NESA approved calculators may be used.
- The NESA Reference Sheet is provided.

Section I

10 marks

Attempt Questions 1 - 10

Allow about 15 minutes for this section

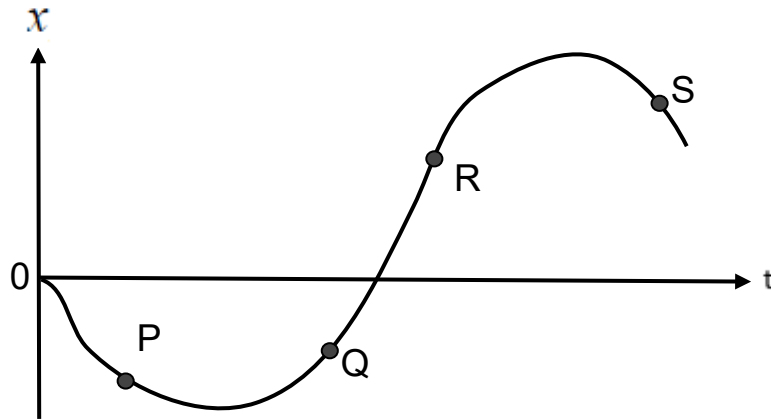
Use the multiple-choice answer sheet for Questions 1 - 10

1. Which of the following is a primitive of $\frac{2}{x} - \sin x$?
 - (A) $2 \ln|x| + \cos x + C$
 - (B) $2 \ln|x| - \cos x + C$
 - (C) $-\frac{2}{x^2} + \cos x + C$
 - (D) $-\frac{2}{x^2} - \cos x + C$
2. What is the value of $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x - 4}$?
 - (A) 0
 - (B) 4
 - (C) 7
 - (D) 10
3. The domain of the function $f(x) = \frac{x}{\sqrt{x+3}}$ is:
 - (A) $x \neq -3$
 - (B) $x > 3$
 - (C) $x > -3$
 - (D) $x \geq -3$
4. Which expression is a term of the geometric series $4x - 8x^2 + 16x^3 - \dots$?
 - (A) $-2048 x^9$
 - (B) $-2048 x^{10}$
 - (C) $2048 x^9$
 - (D) $2048 x^{10}$
5. What is the equation of the tangent to the curve $y = \sin x$ at the point $(\pi, 0)$?
 - (A) $y = 0$
 - (B) $x - y + \pi = 0$
 - (C) $x - y - \pi = 0$
 - (D) $x + y - \pi = 0$

6. What is the value of k if the equation $3x^2 + kx - 4 = 0$ has -2 as one of its roots?

- (A) $k = 4$
- (B) $k = 8$
- (C) $k = -4$
- (D) $k = -8$

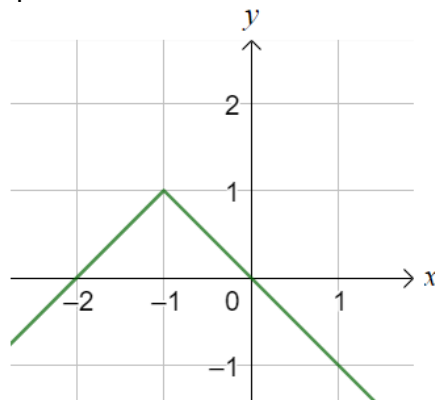
7. The displacement-time graph of a particle moving in a straight line is as shown.



At which point is the velocity negative and the particle is slowing down?

- (A) P
- (B) Q
- (C) R
- (D) S

8. What is the equation of the graph below?

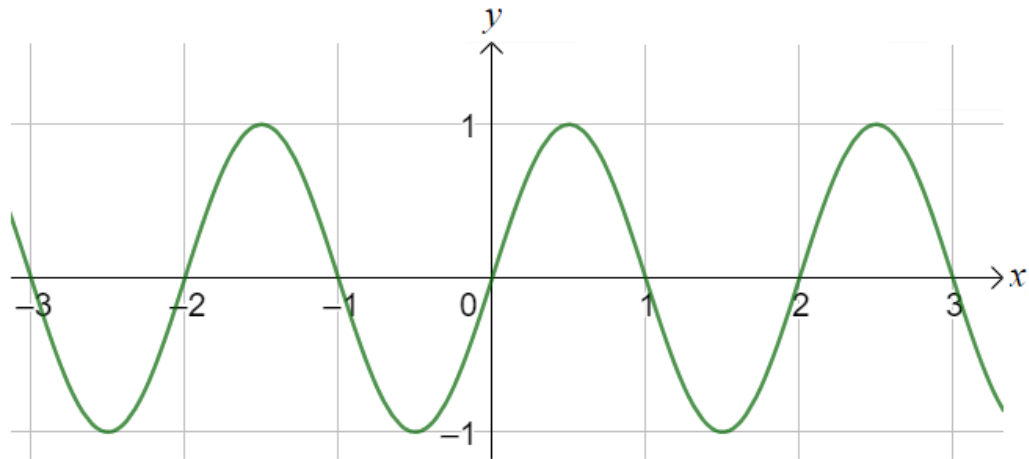


- (A) $y = |x - 1| - 1$
- (B) $y = |x + 1| - 1$
- (C) $y = 1 - |x - 1|$
- (D) $y = 1 - |x + 1|$

9. Which equation represents an odd function?

- (A) $y = \sin x + \cos x$
- (B) $y = \sin x - \cos x$
- (C) $y = \tan x + \sec x$
- (D) $y = \tan x + \operatorname{cosec} x$

10. The graph of the function $y = \sin(\pi x)$ is given below.



How many solutions does $2\sin(\pi x) + x = 0$ have?

- (A) 5
- (B) 3
- (C) 1
- (D) None of the above

End of Section I

Section II

90 marks

Attempt Questions 11 - 16

Allow about 2 hours and 45 minutes for this section

Answer each question in a separate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)	Start a new writing booklet	Marks
(a) Factorise $4x^2 + 11x - 3$		1
(b) If $\frac{8}{5 - \sqrt{2}}$ equals $a + b\sqrt{2}$. Find the values of a and b .		2
(c) Solve $ 2x - 1 = 5$		2
(d) Find the exact value of $\tan \frac{\pi}{6} + \sec \frac{\pi}{4}$		2
(e) For the arithmetic sequence 31, 27, 23, 19,		
(i) What is the 30th term?		1
(ii) Find the sum of the first 50 terms.		1
(f) The equation $2x^2 + 5x - 4 = 0$ has roots α and β . Find the values of		
(i) $\alpha + \beta$		1
(ii) $\alpha\beta$		1
(iii) $(\alpha - 1)(\beta - 1)$		1
(g)		
(i) Sketch the graph of $y = x^2 - 9$, showing all intercepts.		2
(ii) Find the area between the curve $y = x^2 - 9$ and the x -axis.		1

End of Question 11

Question 12 (15 marks)

Start a new writing booklet

Marks

(a) For the parabola $x^2 - 6x - 8y - 31 = 0$,

(i) Show that $x^2 - 6x - 8y - 31 = 0$ can be written as

1

$(x - 3)^2 = 8(y + 5)$ by completing the square.

(ii) Find the focal length.

1

(iii) Find the coordinates of the focus.

1

(iv) Find the equation of the directrix.

1

(v) Find the equation of the normal to the parabola at the point (11,3).

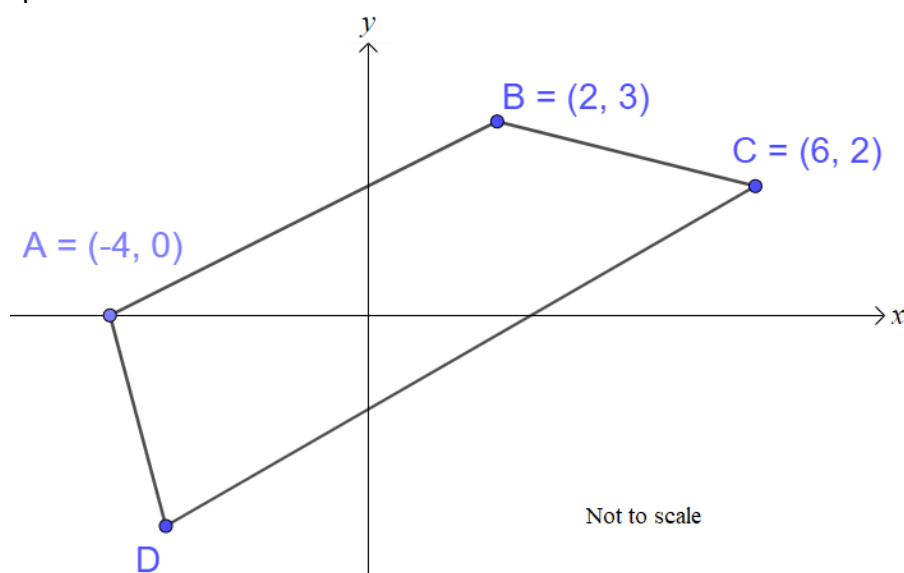
2

(b) The graph of the function $y = f(x)$ passes through the point (1,6).

2

If $f'(x) = 4x - 5$, find $f(x)$.

(c) In the diagram, the points $A(-4,0)$, $B(2,3)$, $C(6,2)$ and D are vertices of a trapezium.



(i) Show that the equation of the line CD is $x - 2y - 2 = 0$.

2

(ii) Given the equation of the line AD is $x + y + 4 = 0$, show that the coordinates of point D are $(-2, -2)$.

2

(iii) Find the length AB. Give your answer in exact form.

1

(iv) Find the perpendicular distance from point A to the line CD.

1

(v) Find the area of the trapezium ABCD if length $CD = \sqrt{80}$.

1**End of Question 12**

Question 13 (15 marks)

Start a new writing booklet

Marks

(a) Solve $4\sin^2 2x = 3$ for $-\pi \leq x \leq \pi$

3

(b) The function $y = f(x)$ is defined by $f(x) = x^3 + 6x^2 + 9x + 1$.

(i) Find any turning points and determine their nature.

3

(ii) Find any point(s) of inflexion.

1

(iii) Sketch the curve, showing all turning points and point(s) of inflexion.

1

(c) The table below gives some values for $f(x) = x \log_e x$

x	1	2	3	4	5
$f(x)$	0		3.33		8.05

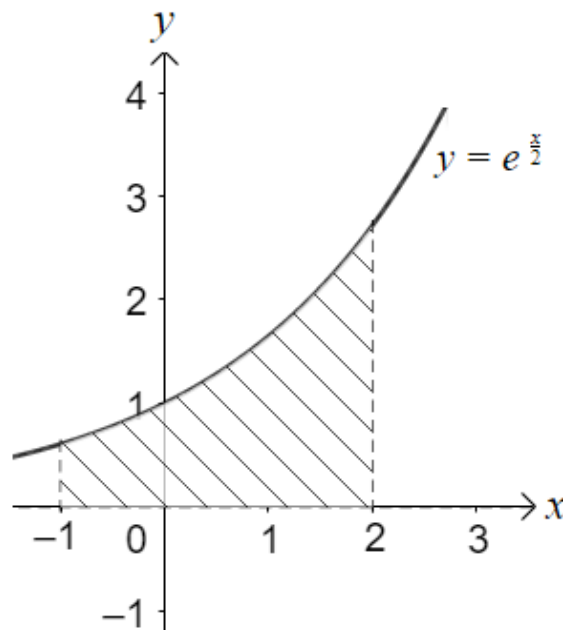
(i) Copy and complete the table of values in your writing booklet.

1

(ii) Use Simpson's Rule with these 5 values to find an approximation of

2

$$\int_1^5 x \log_e x \, dx \quad \text{correct to 2 decimal places.}$$

(d) The region bounded by the graph of $y = x^2$, the x -axis and $x = 2$ is rotated about the x -axis. Find the volume of the solid of revolution formed.**2**(e) The graph of $y = e^{\frac{x}{2}}$ is shown below.**2**

The shaded region is bounded by $y = e^{\frac{x}{2}}$, the x -axis, the lines $x = -1$ and $x = 2$. Find the area of the shaded region, correct to 2 decimal places.

End of Question 13

Question 14 (15 marks)

Start a new writing booklet

Marks

- (a) Mary borrows \$50 000 at 6% compound interest per annum from her bank. She makes \$4000 repayments each year. The amount owed at the end of the n th year is $\$A_n$.

(i) Show that $A_2 = 50\,000(1.06)^2 - (4000(1.06) + 4000)$

1

(ii) Show that $A_n = 50\,000(1.06)^n - \frac{4000(1.06^n - 1)}{0.06}$

2

- (iii) Find the amount of money Mary owes at the end of 12 years. Give your answer correct to the nearest dollar.

1

- (b) If $\log_m a = 1.2$ and $\log_m b = 0.5$, find the value of

(i) $\log_m(ab)$

1

(ii) $\log_m \left(\frac{b^3}{a}\right)$

2

- (c)

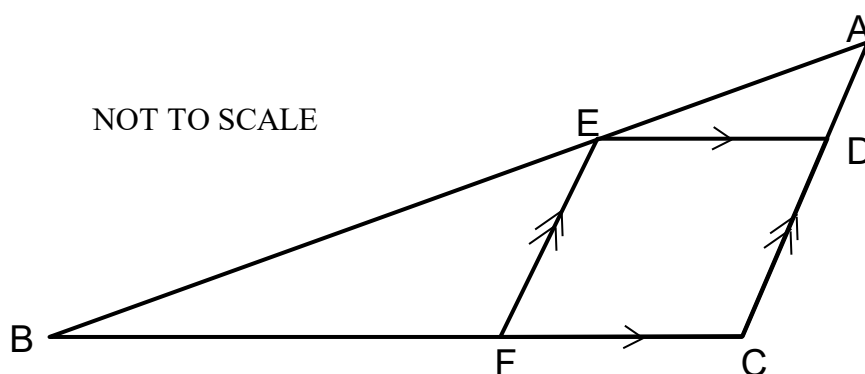
(i) Show that $\frac{1}{2x-3} - \frac{1}{2x+3} = \frac{6}{4x^2-9}$

1

(ii) Hence find $\int \frac{dx}{4x^2-9}$ and fully simplify your answer.

3

- (d)



In the diagram above, CDEF is a rhombus inscribed inside the triangle ABC.

- (i) Prove that $\triangle ADE$ is similar to $\triangle EFB$.

2

- (ii) Let the side of the rhombus be x units, length $BC = 12$ units and $AD = 2$ units. Find the value of x .

2**End of Question 14**

Question 15 (15 marks)

Start a new writing booklet

Marks

- (a) A tank initially containing 500 litres of water, is leaking.

The rate at which water is leaking from the tank is given by $\frac{dV}{dt} = -\frac{1600}{(t+4)^2}$

where V is the volume of water remaining in the tank and t is the time in hours after the water started to leak.

- (i) Find the volume of the water in the tank after 20 hours.
(ii) How much water will be left in the tank eventually?

3

1

- (b) A number of frogs were introduced to an island. The population of frogs after t months is given by $P(t) = Ae^{kt}$, where A and k are constants.

- (i) Show that $P(t)$ satisfies $\frac{dP}{dt} = kP$.

1

- (ii) Five months after the frogs were introduced, their population is now 80% **more** than the initial population.

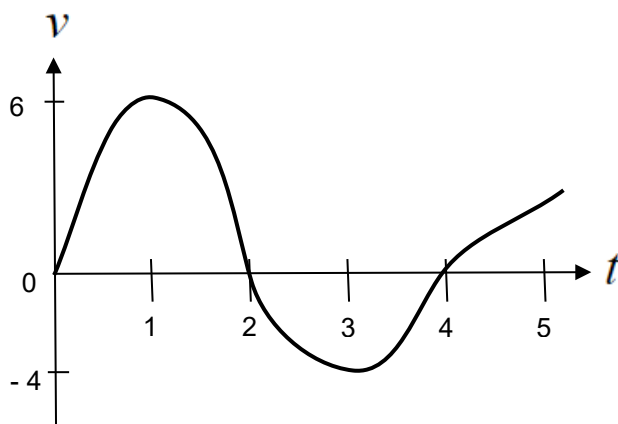
2

Show that $k = 0.1176$ correct to 4 significant figures.

- (iii) After how many months will the number of frogs first exceed one hundred times its initial population?

2

- (c)



The graph above shows the velocity v , in m/s of a particle after t seconds moving in a straight line.

- (i) What is the acceleration when $t = 1$ second?
(ii) When does the particle first change direction?
(iii) What does the area under the curve from $t = 0$ to $t = 2$ represent?

1

1

1

- (d) A particle that is initially at rest at the origin starts moving along the x -axis.

The velocity of the particle at time t is given by $\dot{x} = 12(t - t^3)$ ms⁻¹.

- (i) Once moving, when does the particle come to rest again?
(ii) Find the distance travelled in the first 2 seconds.

1

2

End of Question 15

Question 16 (15 marks)

Start a new writing booklet

Marks(a) Differentiate with respect to x

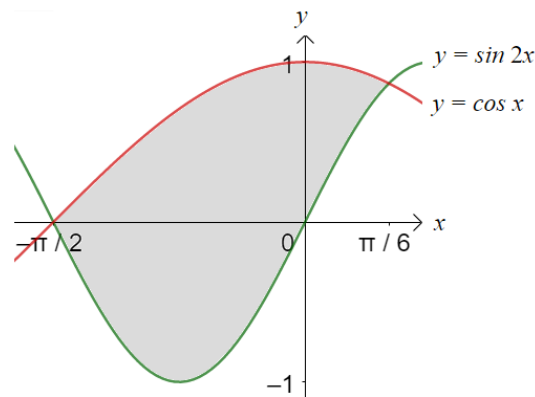
(i) $\tan \frac{x}{3}$

1

(ii) $\sin^4 x$

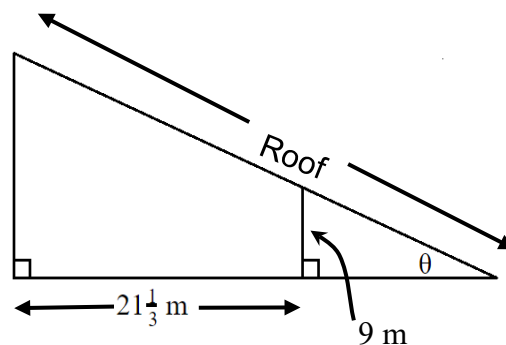
1(b) Find $\int \sec^2 \left(\frac{x-3}{2} \right) dx$ **1**(c) (i) Given $y = \sin x^2$, find y' **1**(ii) Hence, or otherwise, determine $\int x \cos x^2 dx$ **2**

(d)

3

The diagram shows the graphs of $y = \cos x$ and $y = \sin 2x$. The two graphs intersect at $x = -\frac{\pi}{2}$ and $x = \frac{\pi}{6}$. Calculate the area of the shaded regions.

(e)



A structure is made of two vertical walls and a roof, making an angle θ with the ground. The horizontal distance between the two walls is $21\frac{1}{3}$ metres and the height of the shorter wall is 9 metres.

(i) Show that the length of the roof is $L = \frac{9}{\sin \theta} + \frac{64}{3 \cos \theta}$ **3**

(ii) Find the length of the shortest roof.

3**End of Paper**

HSC Mathematics

AP4 2018

NAME _____

CLASS 12 MA _____ 12 MTX _____

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

Question	Marks
1-10	/ 10
11	/ 15
12	/ 15
13	/ 15
14	/ 15
15	/ 15
16	/ 15
Total	/ 100

1) A 2) C 3) C 4) B 5) D 6) A 7) A 8) D 9) D 10) A

11) a) $4x^2 + 11x - 3$

$$= (4x - 1)(x + 3)$$

b) $\frac{8(5 + \sqrt{2})}{(5 - \sqrt{2})(5 + \sqrt{2})}$

$$= \frac{40 + 8\sqrt{2}}{23}$$

$$= \frac{40}{23} + \frac{8\sqrt{2}}{23}$$

$$a = \frac{40}{23}, b = \frac{8}{23}$$

c) $2x - 1 = \pm 5$

$$2x = -4 \quad \text{or} \quad 2x = 6$$

$$x = -2$$

$$x = 3$$

d) $\frac{1}{\sqrt{3}} + \sqrt{2}$

$$= \frac{\sqrt{3}}{3} + \sqrt{2}$$

$$= \frac{\sqrt{3} + 3\sqrt{2}}{3}$$

e) i) $a = 31, d = 27 - 31$
 $d = -4$

$$T_n = a + (n-1)d$$

$$T_{30} = 31 + (30-1)(-4)$$

$$= -85$$

ii) $S_n = \frac{n}{2} [2a + (n-1)d]$

$$S_{50} = \frac{50}{2} [2(31) + (50-1)(-4)]$$

$$= -3350$$

$$a=2, b=5, c=-4$$

$$f) \alpha + \beta = -\frac{b}{a}$$

$$= -\frac{5}{2}$$

$$ii) \alpha\beta = \frac{c}{a}$$

$$= \frac{-4}{2}$$

$$= -2$$

$$iii) (\alpha-1)(\beta-1)$$

$$= \alpha\beta - \alpha - \beta + 1$$

$$= \alpha\beta - (\alpha + \beta) + 1$$

$$= -2 - \left(-\frac{5}{2}\right) + 1$$

$$= \frac{3}{2}$$

$$9) i) y = x^2 - 9$$

$$y\text{-intercept: } x=0$$

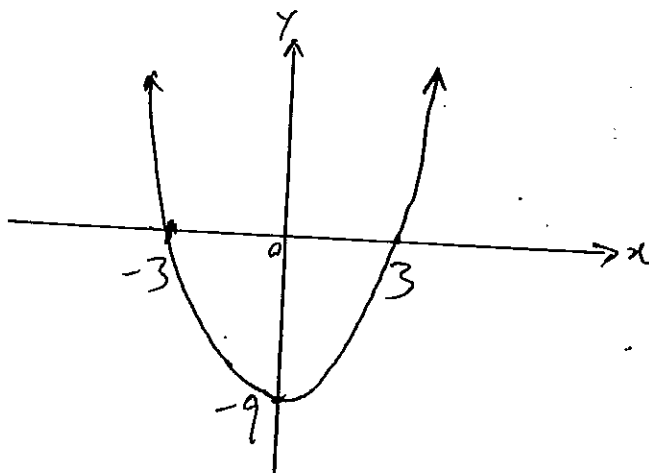
$$y = -9$$

$$x\text{-intercepts: } y=0$$

$$0 = x^2 - 9$$

$$x^2 = 9$$

$$x = \pm 3$$



Alternatively,

$$A = 2 \left| \int_0^3 x^2 - 9 dx \right|$$

$$= 2 \left| \left[\frac{x^3}{3} - 9x \right]_0^3 \right|$$

$$= 2 \left| [9 - 27] \right|$$

$$= 36 \text{ units}^2$$

$$ii) A = \left| \int_{-3}^3 x^2 - 9 dx \right|$$

$$= \left| \left[\frac{x^3}{3} - 9x \right]_{-3}^3 \right|$$

$$= 36 \text{ units}^2$$

$$12 \text{ a) i) } x^2 - 6x - 8y - 31 = 0$$

$$x^2 - 6x + 9 - 9 = 8y + 31$$

$$(x-3)^2 = 8y + 40$$

$$(x-3)^2 = 8(y+5) \rightarrow \text{GIVEN}$$

$$\text{ii) } 8 = 4a$$

$$a = 2$$

$$\text{focal length} = 2$$

$$\text{iii) Focus} = (3, -3)$$

$$\text{iv) Eqn of directrix:}$$

$$\text{v) } y = \frac{(x-3)^2}{8} - 5$$

$$\frac{dy}{dx} = \frac{x-3}{4}$$

$$\text{At } x=11,$$

$$\frac{dy}{dx} = \frac{11-3}{4}$$

$$= 2$$

$$m_{\text{normal}} = -\frac{1}{2}$$

$$\text{Eqn of normal:}$$

$$y-3 = -\frac{1}{2}(x-11)$$

$$2y-6 = -x+11$$

$$x+2y-17=0$$

$$\text{b) } f'(x) = 4x - 5$$

$$f(x) = \int 4x - 5$$

$$f(x) = 2x^2 - 5x + C$$

$$6 = 2(1)^2 - 5(1) + C$$

$$C = 9$$

$$\therefore f(x) = 2x^2 - 5x + 9$$

$$\left. \begin{array}{l} y = -7 \\ \text{OR } y = \frac{x^2 - 6x + 9 - 40}{8} \\ = \frac{x^2 - 6x - 31}{8} \end{array} \right\} \text{cfm}$$

$$y' = \frac{x}{4} - \frac{3}{4}$$

$$\text{at } x=11, m_{\text{tangent}} = 2$$

$$c) i) m_{CD} = m_{AB}$$

$$\frac{y-2}{x-6} = \frac{3}{2-(-4)}$$

$$\frac{y-2}{x-6} = \frac{1}{2}$$

$$2y-4 = x-6$$

$$x-2y-2=0 \rightarrow \text{GIVEN}$$

$$ii) \text{ Line CD: } x-2y-2=0 \quad \text{--- (1)}$$

$$\text{Line AD: } x+y+4=0 \quad \text{--- (2)}$$

$$(2)-(1): 3y+6=0$$

$$y=-2$$

$$\text{Sub } y=-2 \text{ into (1)}$$

$$x-2(-2)-2=0$$

$$x=-2$$

$$D = (-2, -2) \rightarrow \text{GIVEN}$$

$$iii) AB = \sqrt{[2-(-4)]^2 + 3^2}$$

$$= \sqrt{45} \text{ units}$$

$$= 3\sqrt{5} \text{ units}$$

$$iv) d = \frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}}$$

$$= \frac{|(1)(-4)+0+(-2)|}{\sqrt{1^2+(-2)^2}}$$

$$= \frac{6}{\sqrt{5}}$$

$$= \frac{6\sqrt{5}}{5} \text{ units}$$

$$v) A = \frac{1}{2} (3\sqrt{5} + \sqrt{80}) \times \frac{6\sqrt{5}}{5}$$

$$= (3\sqrt{5} + 4\sqrt{5}) \times \frac{3\sqrt{5}}{5}$$

$$= \frac{7\sqrt{5} \times 3\sqrt{5}}{5}$$

$$= 21 \text{ units}^2$$

$$\text{OR } m_{AB} = \frac{3-0}{2-(-4)} = \frac{3}{6} = \frac{1}{2}$$

equation of CD // to AB

$$(y-2) = \frac{1}{2}(x-6)$$

$$2y-4 = x-6$$

$$0 = x-2y-2 \text{ GIVEN}$$

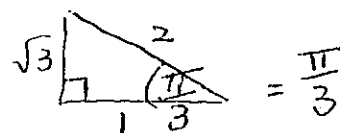
$$1.5 a) 4 \sin^2 2x = 3 \quad -\pi \leq x \leq \pi$$

$$\sin 2x = \pm \frac{\sqrt{3}}{2} \quad -2\pi \leq 2x \leq 2\pi$$

$$2x = +\frac{\pi}{3}, +\frac{2\pi}{3}, +\frac{4\pi}{3}, +\frac{5\pi}{3}$$

$$x = +\frac{\pi}{6}, +\frac{\pi}{3}, +\frac{2\pi}{3}, +\frac{5\pi}{6}$$

$$\pm 30^\circ, \pm 60^\circ, \pm 120^\circ, \pm 150^\circ$$



$$b) f(x) = x^3 + 6x^2 + 9x + 1$$

$$i) f'(x) = 3x^2 + 12x + 9$$

$$f''(x) = 6x + 12$$

$$\text{Stationary points: } f'(x) = 0$$

$$0 = 3x^2 + 12x + 9$$

$$0 = x^2 + 4x + 3$$

$$0 = (x+1)(x+3)$$

$$x = -1 \quad \text{or} \quad x = -3$$

$$y = -3$$

$$y = 1$$

$$f''(-1) = 6(-1) + 12$$

$$= -6 + 12$$

$$= 6$$

$$> 0$$

$$f''(-3) = 6(-3) + 12$$

$$= -18 + 12$$

$$= -6$$

$$< 0$$

$(-1, -3)$ is a ^{rel} minimum and $(-3, 1)$ is a ^{rel} maximum.

$$ii) \text{ Possible pt of inflexion: } f''(x) = 0$$

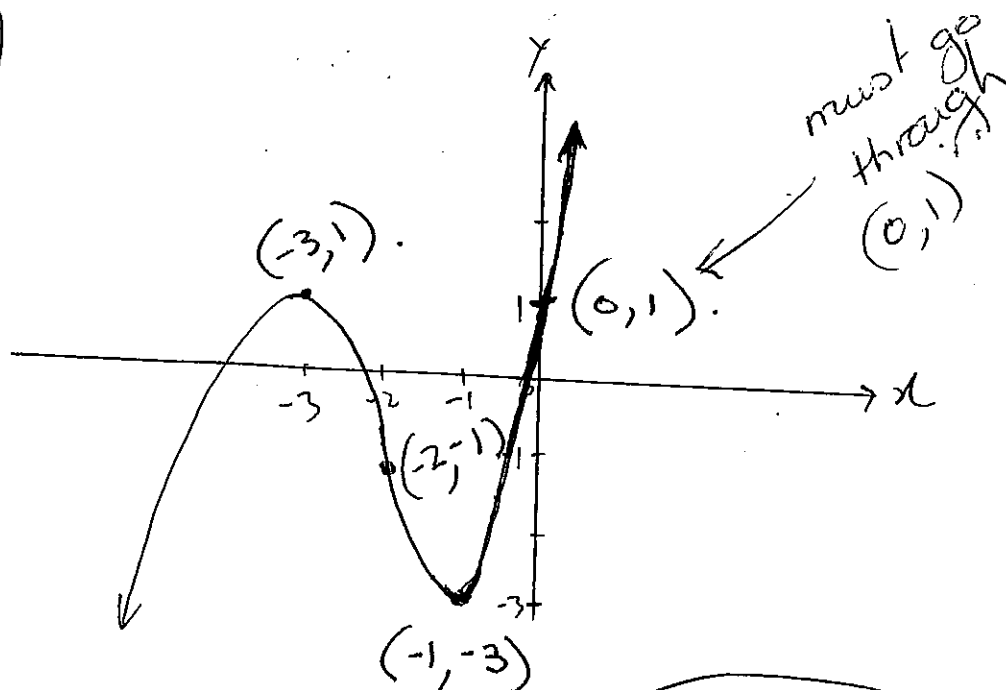
$$0 = 6x + 12$$

$$x = -2, y = -1$$

x	-3	-2	-1
$f''(x)$	-6	0	6

Concavity changes. $\therefore (-2, -1)$ is a point of inflexion.

iii)



c) i)

	y_0	y_1	y_2	y_3	y_4
x	1	2	3	4	5
$f(x)$	0	1.39	3.33	5.55	8.05

3.3

$$ii) \int_1^5 x \ln x \, dx = \int_1^3 x \ln x \, dx + \int_3^5 x \ln x \, dx$$

$$\approx \frac{3-1}{6} [0 + 4(1.39) + 3.33] + \frac{5-3}{6} [3.33 + 4(5.55) + 8.05]$$

OR $\frac{1}{3} \times \frac{5-1}{4} = 14.16$
 $[0 + 8.05 + 2(3.33) + 4(1.39 + 5.55)]$ (1 mark for 28.31 (chr

d) $V = \pi \int_0^2 y^2 \, dy$

$$= \pi \int_0^2 x^4 \, dx$$

$$= \pi \left[\frac{x^5}{5} \right]_0^2$$

$$= \frac{32\pi}{5} \text{ units}^3 \text{ or } 6.4\pi$$

$$V = \pi \int_0^2 x^2 \, dx$$

$$= \frac{8\pi}{3}$$

OR 20.1

e) $A = \int_{-1}^2 e^{\frac{x}{2}} \, dx$

$$= 2 \left[e^{\frac{x}{2}} \right]_{-1}^2$$

$$= 2 \left[e - e^{-\frac{1}{2}} \right] =$$

$$\approx 0.79 \text{ (2 d.p.)}$$

$$= 4.22$$

Question 14

$$\begin{aligned} \text{a) i) } A_1 &= 50000 (1 + 0.06) - 4000 \\ &= 50000 (1.06) - 4000 \end{aligned}$$

$$A_2 = A_1 \times 1.06 - 4000$$

$$\begin{aligned} A_2 &= [50000 (1.06) - 4000] \times 1.06 - 4000 \\ &= 50000 (1.06)^2 - 4000 (1.06) - 4000 \\ &= 50000 (1.06)^2 - (4000 (1.06) + 4000) \quad \text{as required.} \end{aligned}$$

$$\begin{aligned} \text{ii) } A_3 &= A_2 (1.06) - 4000 \\ &= [50000 (1.06)^2 - (4000 (1.06) + 4000)] (1.06) - 4000 \\ &= 50000 (1.06)^3 - 4000 (1.06)^2 - 4000 (1.06) - 4000 \\ &= 50000 (1.06)^3 - 4000 (1.06^2 + 1.06 + 1) \end{aligned}$$

$$A_n = 50000 (1.06)^n - 4000 \underbrace{(1.06^{n-1} + 1.06^{n-2} + \dots + 1.06 + 1)}_{\text{is a GP with } a=1 \quad r=1.06 \quad n \text{ terms}}$$

$$\begin{aligned} A_n &= 50000 (1.06)^n - 4000 \left[1 \times \frac{1.06^n - 1}{1.06 - 1} \right] \\ &= 50000 (1.06)^n - \frac{4000 (1.06^n - 1)}{0.06} \quad \text{as required.} \end{aligned}$$

$$\text{iii) } A_{12} = 50000 (1.06)^{12} - \frac{4000 (1.06^{12} - 1)}{0.06}$$

$$\begin{aligned} &= \$33130.06 \\ &= \$33130 \quad (\text{nearest dollar}) \end{aligned}$$

$$\begin{aligned}
 \text{b) i) } \log_m(ab) &= \log_m a + \log_m b \\
 &= 1.2 + 0.5 \\
 &= 1.7
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } \log_m \frac{b^3}{a} &= \log_m b^3 - \log_m a \\
 &= 3 \log_m b - \log_m a \\
 &= 3(0.5) - 1.2 \\
 &= 0.3
 \end{aligned}$$

$$\begin{aligned}
 \text{c) i) } \frac{1}{2x-3} - \frac{1}{2x+3} &= \frac{1(2x+3) - 1(2x-3)}{(2x-3)(2x+3)} \\
 &= \frac{2x+3-2x+3}{4x^2-9} \\
 &= \frac{6}{4x^2-9}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } \int \frac{dx}{4x^2-9} &= \frac{1}{6} \int \frac{6}{4x^2-9} dx \\
 &= \frac{1}{6} \int \frac{1}{2x-3} - \frac{1}{2x+3} dx \\
 &= \frac{1}{6} \times \frac{1}{2} \int \frac{2}{2x-3} - \frac{2}{2x+3} dx \\
 &= \frac{1}{12} [\ln |2x-3| - \ln |2x+3|] + C \\
 &= \frac{1}{12} \ln \left| \frac{2x-3}{2x+3} \right| + C
 \end{aligned}$$

In $\triangle ADE$ and $\triangle EFB$

d) i) $\angle DEA = \angle FBE$ (corresponding angles on $ED \parallel BC$ are equal)

$\angle DAE = \angle FEB$ (corresponding angles on $AC \parallel EF$ are equal)

$\therefore \triangle ADE \parallel \triangle EFB$ (equiangular)

NOTE: Above is the most efficient solution but other solutions supported by valid reasons were accepted.

ii) $\frac{AD}{DE} = \frac{EF}{BF}$ (matching sides in $\triangle ADE \parallel \triangle EFB$ are in proportion)

$$\frac{2}{x} = \frac{x}{12-x}$$

$$24 - 2x = x^2$$

$$x^2 + 2x - 24 = 0$$

$$(x+6)(x-4) = 0$$

$$x = -6, 4$$

But x is a length $\therefore x > 0$ Hence $x = 4$.

Note: Students who used the ratio

$$\frac{2}{x} = \frac{2+x}{12}$$

gained 1 mark if they correctly solved for x , but did not gain credit as they had not established $\triangle ADE \parallel \triangle ACF$

$$15) a) i) \frac{dV}{dt} = -\frac{1600}{(t+4)^2}$$

$$= -1600(t+4)^{-2}$$

$$V = \int -1600(t+4)^{-2} dt$$

$$= 1600(t+4)^{-1} + C$$

$$V = \frac{1600}{t+4} + C$$

When $t=0$, $V=500$.

$$500 = \frac{1600}{4} + C$$

$$C = 100$$

$$\therefore V = \frac{1600}{t+4} + 100$$

When $t=20$,

$$V = \frac{1600}{20+4} + 100 \quad \text{or } \frac{500}{3} \text{ L}$$

$$= 166\frac{2}{3} \text{ litres}$$

ii) When $t \rightarrow \infty$,

$$V \rightarrow 100 \quad \text{as } \frac{1600}{t+4} \rightarrow 0$$

The volume will eventually be reduced to 100L.

b) i) $P = Ae^{kt}$

$$\frac{dP}{dt} = Ake^{kt}$$

$$= k(Ae^{kt})$$

$$= kP$$

ii) $P = 1.8A$ when $t=5$

$$1.8A = Ae^{5k}$$

$$\ln 1.8 = 5k$$

$$k = \frac{\ln 1.8}{5}$$

$$\approx 0.1176 \quad (4 \text{ sig. figs.})$$

$$\text{iii) } P = Ae^{0.1176t}$$

$$P > 100A$$

$$Ae^{0.1176t} > 100A$$

$$0.1176t > \ln 100$$

$$t > \frac{\ln 100}{0.1176}$$

$$t > 39.1596 \dots$$

$$t = 40$$

It will take 40 months.

39.16 or 39.2 also accepted.

$$\text{c) i) } a = 0.$$

$$\text{ii) } t = 2$$

iii) It represents the distance travelled ^{or displacement} in the first two seconds.

$$\text{d) i) } \dot{x} = 0$$

$$0 = 12(t - t^3)$$

$$0 = t(1 - t^2)$$

$$0 = t(1 + t)(1 - t)$$

$$t = 0, -1, 1$$

The particle comes to rest again after 1 second.

$$\text{ii) Distance travelled} = \left| \int_0^1 12(t - t^3) dt \right| + \left| \int_1^2 12(t - t^3) dt \right|$$

$$= \left| 12 \left[\frac{t^2}{2} - \frac{t^4}{4} \right]_0^1 \right| + \left| 12 \left[\frac{t^2}{2} - \frac{t^4}{4} \right]_1^2 \right|$$

$$= 12 \left(\frac{1}{2} - \frac{1}{4} \right) + \left| 12 \left[(2 - 4) - \left(\frac{1}{2} - \frac{1}{4} \right) \right] \right|$$

$$= 3 + 27$$

$$= 30 \text{ m}$$

or
Finding displacement function and constant of integration.

$$16) a) i) \frac{d}{dx} \left(\tan \frac{x}{3} \right) = \frac{1}{3} \sec^2 \left(\frac{x}{3} \right)$$

$$ii) \frac{d}{dx} (\sin^4 x) = 4 \sin^3 x \cos x$$

$$b) \int \sec^2 \left(\frac{x-3}{2} \right) dx = 2 \tan \left(\frac{x-3}{2} \right) + C$$

$$c) i) y = \sin x^2$$

$$y' = 2x \cos x^2$$

$$ii) \int x \cos x^2 dx = \frac{1}{2} \int 2x \cos x^2 dx$$

$$= \frac{1}{2} \sin x^2 + C$$

$$2 \int x \cos x^2 dx$$

$$\frac{1}{2} \sin x^2 + C$$

$$d) A = \int_{-\frac{\pi}{2}}^{\frac{\pi}{6}} \cos x - \sin 2x dx$$

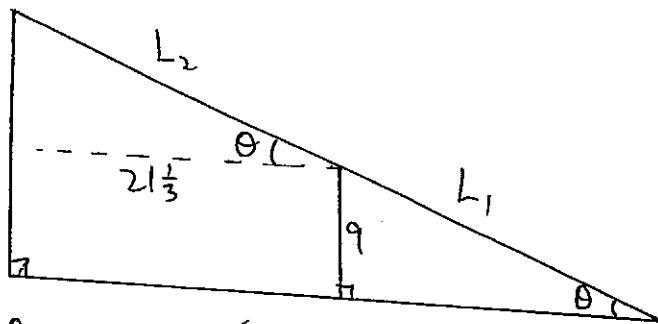
$$= \left[\sin x + \frac{\cos 2x}{2} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{6}}$$

$$= \left[\left(\sin \frac{\pi}{6} + \frac{\cos \frac{\pi}{3}}{2} \right) - \left(\sin \left(-\frac{\pi}{2} \right) + \frac{\cos(-\pi)}{2} \right) \right]$$

$$= \frac{1}{2} + \frac{1}{4} - \left(-1 - \frac{1}{2} \right)$$

$$= 2 \frac{3}{4} \text{ units}^2$$

e) i)



$$\sin \theta = \frac{9}{L_1} \quad \cos \theta = \frac{21\frac{1}{3}}{L_2}$$

$$L_1 = \frac{9}{\sin \theta}$$

$$L_2 = \frac{21\frac{1}{3}}{\cos \theta}$$

$$L_2 = \frac{64}{3 \cos \theta}$$

$$L = L_1 + L_2$$

$$= \frac{9}{\sin \theta} + \frac{64}{3 \cos \theta}$$

ii)

$$L = 9(\sin \theta)^{-1} + \frac{64}{3}(\cos \theta)^{-1}$$

$$\frac{dL}{d\theta} = -9(\sin \theta)^{-2} \cos \theta - \frac{64}{3}(\cos \theta)^{-2}(-\sin \theta)$$

$$= -\frac{9 \cos \theta}{\sin^2 \theta} + \frac{64 \sin \theta}{3 \cos^2 \theta}$$

$$\frac{dL}{d\theta} = 0, \quad 0 = -\frac{9 \cos \theta}{\sin^2 \theta} + \frac{64 \sin \theta}{3 \cos^2 \theta}$$

$$27 \cos^3 \theta = 64 \sin^3 \theta$$

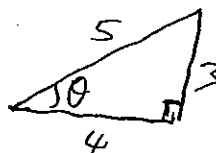
$$\tan^3 \theta = \frac{27}{64}$$

$$\tan \theta = \frac{3}{4}$$

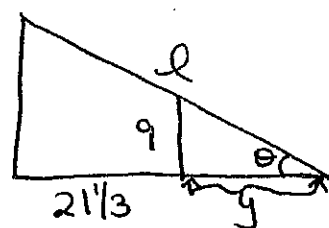
$$\theta = \tan^{-1}\left(\frac{3}{4}\right)$$

θ	35°	$\tan^{-1}\left(\frac{3}{4}\right)$	40°
$\frac{dL}{d\theta}$	-0.921	0	1.620

Minimum $L = \frac{9}{\sin \theta} + \frac{64}{3 \cos \theta}$
 $= 9 \times \frac{5}{3} + \frac{64 \times 5}{3 \times \frac{4}{5}}$



$$\rightarrow = 41\frac{2}{3} \text{ m}$$



$$\tan \theta = \frac{9}{y}$$

$$y = \frac{9}{\tan \theta}$$

$$\cos \theta = \frac{21\frac{1}{3} + y}{l}$$

$$l = \frac{21\frac{1}{3} + y}{\cos \theta}$$

$$l = \frac{1}{\cos \theta} \left[21\frac{1}{3} + y \right]$$

$$= \frac{1}{\cos \theta} \left[\frac{64}{3} + \frac{9}{\tan \theta} \right]$$

$$= \frac{64}{3 \cos \theta} + \frac{9}{\sin \theta \cos \theta}$$

$$= \frac{64}{3 \cos \theta} + \frac{9}{\sin \theta}$$